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1 Introduction

1.1 Background

The transition from extraterrestrial exploration to permanent colonization represents the next "Great Filter" for human civilization. As we approach the mid-21st century, the ambition to establish a self-sustaining lunar colony for 100,000 inhabitants by 2050 has moved from science fiction to engineering reality.

However, it presents an unprecedented logistical challenge: transporting 100 million metric tons of infrastructure materials. This endeavor exposes a fundamental trade-off between traditional chemical rocketry—characterized by high speed but prohibitive costs and emissions—and the proposed Space Elevator System, which offers sustainable, low-cost transit but faces capacity constraints and significant upfront deployment risks.

The Moon Colony Management (MCM) Agency must navigate this complex landscape to optimize construction schedules and budgets. The core challenge lies in balancing the immediate efficiency of rockets against the long-term sustainability of space elevators, while simultaneously ensuring system resilience against catastrophic failures and minimizing the environmental footprint on Earth.

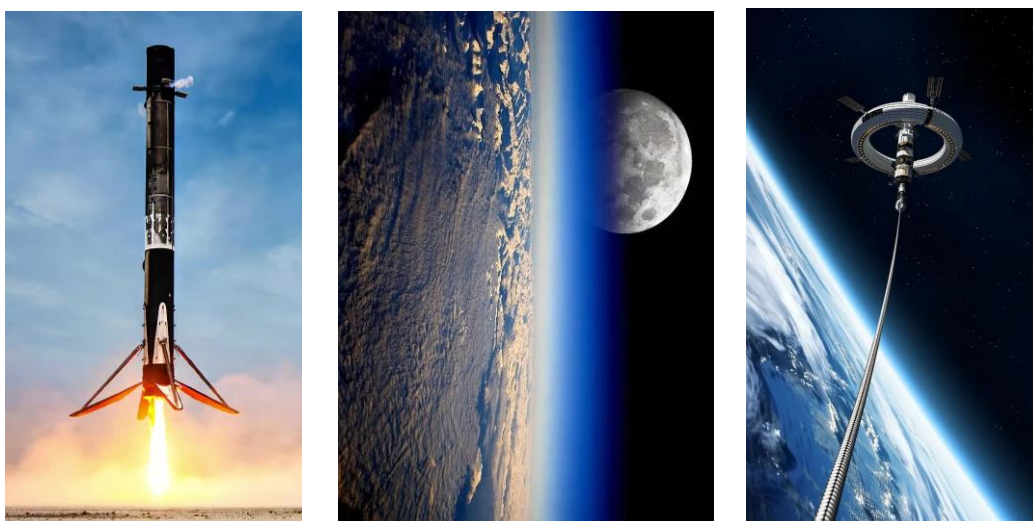


Figure 1: The Falcons, The Moon & The elevator

1.2 Problem Restatement

To assist the MCM Agency in formulating a scientifically grounded roadmap, we model the Earth-Moon logistics network as a complex, dynamic system. Our research objectives are decomposed into four distinct but interconnected tasks:

- **Task 1: Strategic Logistics Planning (Phase I).** Develop a multi-objective optimization model to determine the optimal transport architecture (Elevator, Rocket, or Hybrid). The objective is to construct a Pareto frontier that minimizes both Total Cost (\$) and Construction Timeline (Years).
- **Task 2: Resilience & Risk Quantification (Phase II).** Assess system vulnerability to stochastic disruptions, specifically rocket launch failures and elevator tether swaying. We aim to design a

"Dual-Buffer" strategy to ensure mission success without unmanageable budget overruns.

- **Task 3: Operational Metabolism (Phase III).** Upon colony completion, focus shifts to sustaining the population. We model the colony's water cycle and biological lock-in to determine the most economically viable "Weekly Pulse" replenishment schedule.
- **Task 4: Environmental Evaluation (Phase IV).** Quantify negative externalities (e.g., atmospheric pollution, space debris) and propose optimization strategies to transition the project from pure engineering efficiency to Sustainable Interstellar Governance.

1.3 Our works

To intuitively reflect our research process, a comprehensive modeling framework is designed as shown in Figure 2. Our work is structured into four core modules:

- Phase I: Dynamic Logistics Optimization. Solving the 100M-ton transportation schedule using a Mixed-Integer Programming (MIP) approach.
- Phase II: System Reliability Model. Assessing the impact of equipment failures and designing resilient response strategies.
- Phase III: Resource Supply Model. Calculating long-term water demand and sustainable supply cycles for the lunar colony.
- Phase IV: Environmental Exteriority Evaluation. Quantifying the environmental footprint of logistics activities based on Life Cycle Assessment (LCA).

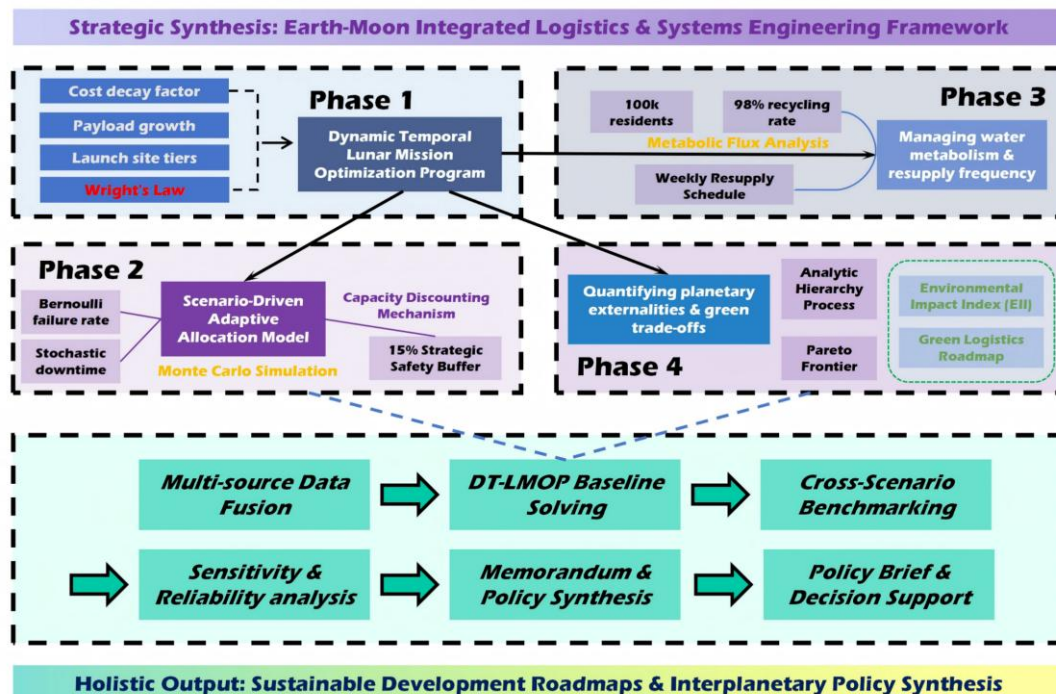


Figure 2: The Framework of Our Work

2 Assumption & Justification

- **Assumption 1: The model utilizes a discrete time horizon with annual accounting periods.**
Justification: Long-term infrastructure projects (Capital Budgeting) are typically evaluated using annual Discounted Cash Flow models, making annual discretization consistent with standard financial reporting. Furthermore, converting continuous dynamics into a discrete-time system reduces computational complexity for century-long planning while retaining the ability to capture macroscopic strategic trends.
- **Assumption 2: Operations follow techno-economic duality and exhibit learning effects.**
Justification: Based on Wright's Law, high-tech manufacturing benefits from "learning by doing," where unit costs decrease as cumulative volume increases. Distinguishing between infrastructure (Fixed Assets) and fuel/transport (Variable Costs) aligns with Cost-Volume-Profit analysis. Assuming linear variable costs allows us to focus on long-term trends rather than short-term step-cost fluctuations.
- **Assumption 3: Annual Supply chain channel selection is driven by pure economic rationality.**
Justification: Under the Rational Choice Theory, decision-makers are assumed to be profit-maximizers who select transport modes solely to minimize Annual Operating Expenses. This creates a "Dual Sourcing" strategy common in supply chain management, utilizing the space elevator as the "efficient source" for stability and rockets as the "responsive source" for resilience.
- **Assumption 4: Inventory dynamics and supply disruption risks are deterministic.**
Justification: Adopting a "Certainty Equivalent" approach avoids the computational burden of stochastic processes in a long-term strategic model. Assuming constant demand and predictable disruption durations (e.g., tether instability) aligns with the premises of the Economic Order Quantity model, allowing for the derivation of clear, optimal safety stock boundaries.
- **Assumption 5: Multi-objective optimization relies on linear weighting and Pareto trade-offs.**
Justification: The Weighted Sum Method is a standard approach in Multi-Attribute Utility Theory, implying that trade-offs between objectives (e.g., cost vs. environment) are compensable. This effectively constructs a Social Welfare Function where weight adjustments represent the changing "shadow prices" policy-makers assign to environmental benefits versus financial costs.
- **Assumption 6: Decision standards are constrained by a bounded strategic time horizon.**
Justification: Behavioral economics suggests that intertemporal choices are subject to heavy discounting; projects exceeding a specific duration suffer from diminishing strategic value regardless of cost efficiency. This constraint aligns mathematical optimization with the managerial concept of "Strategic Windows," restricting solutions to a socially and politically feasible timeframe.

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Major notations

<i>Notation</i>	<i>Definition</i>	<i>Unit</i>
M_{total}	Total demand of material quality	metric-ton
$C_{elevator}$	Unit cost of space elevator from ground to summit anchor	USD
$Height$	Height of space elevator	km
$Cap_{rocket,i}$	Payload capacity per rocket at ground launch sites	metric-ton
$C_{rocket1}$	Unit transport cost per effective payload from ground	USD/kg
$\eta_{rocket2}$	Payload factor from apex anchor to Moon-launch rocket	%
$C_{rocket2}$	Unit transport cost per effective payload from apex anchors	USD/kg
$m_{rocket,i}$	Number of rockets launched per year from ground sites	metric-ton
$Freq_{max,i}$	Maximum annual launches at ground sites in 2050	flts/year
$Cap_{elevator}$	Space elevator transportation capacity limit	metric-ton
N_{ports}	Number of Galactic Harbours	count
N_{site}	Number of ground launch sites	count
$T_{construct}$	Total construction time	year
C_{total}	Total cost	USD
α	Ratio of construction time to cost in the objective function	%
γ	Spacecraft/fuel optimization ratio	%
P_{fail_rocket}	Rocket failure probability	%
P_{fail_ele}	Space elevator failure/ downtime probability	%

4 Model Preparation

4.1 Data Overview

Since the problem statement does not provide specific constants for all necessary physical and economic variables, we collected external data and established reasonable assumptions to define the model's parameters. We focused on extrapolating current technological trends to the year 2050. The derivation and definition of these key constants are detailed below.

4.2 Data Collections

Table 2: Data and Database Websites

<i>Data</i>	<i>Description</i>
Falcon Heavy basic parameters	https://www.spacex.com/vehicles/falcon-heavy .
Rocket launch cost	https://ourworldindata.org/grapher .
Per capita municipal water uses per year	https://data.apps.fao.org .
Population of each country	https://data.worldbank.org .
Per capita required arable land	https://lsda.jsc.nasa.gov .

4.3 Vital Parameter Estimations

We define $Cap_{rocket,i}$ as the effective payload capacity of a rocket launched from a ground-based site. Assuming a baseline capacity of 150 metric tons at the equator (as defined in the problem), we model the impact of latitude while ignoring transient factors like weather. We adopt a linearized reduction model where payload capacity decreases by 0.5% for every degree of latitude deviation from the equator:

$$Cap_{rocket,i} = Cap_{base} \times (1 - 0.005 \times |Latitude_i|)$$

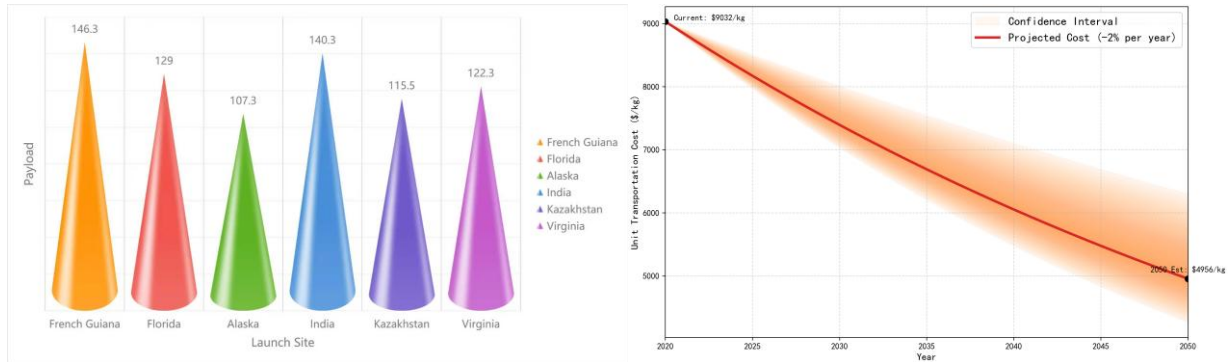


Figure 3: Estimated payload and rocket costs for each launch site

To estimate the unit cost of transporting mass from Earth to the Moon via traditional rockets, we utilized 2018 data for the Falcon Heavy launch vehicle [1][2]. However, lacking official data for direct Lunar Transfer Orbit missions, we applied a linear interpolation model based on Payload-to-Energy ratios from Low Earth Orbit and Geostationary Transfer Orbit data.

We further accounted for the additional energy required for a soft lunar landing. **Using the Tsiolkovsky rocket equation** with a correction parameter, and assuming a yearly technological cost reduction of 2% (cost becomes 98% of the previous year).

In this way, we estimate 2050-unit cost as: $C_{rocket1} \approx 4956\$/kg$.

We estimate $C_{elevator}$ based on the specific orbital energy difference (ΔE) between the surface and the Apex Anchor. Assuming an energy conversion efficiency η and electricity price P_{elec} , the cost is derived as follows:

$$C_{elevator} = \eta \cdot P_{elec} \cdot \Delta E \approx 1.33\$/kg$$

To estimate the unit cost of transporting mass from Apex Anchor to the Moon via rockets, we assume that the transport cost is derived primarily from the energy input required.

Regarding launches from the Apex Anchor, we acknowledge that the high tangential velocity at the anchor point creates a complex orbital transfer scenario. However, **for the 2050 timeframe, we assume mature orbital transfer technology allows for efficient deceleration.** Considering this, we introduce a sensitivity parameter γ , representing the fuel consumption ratio of an Apex-launch versus a traditional Earth-launch. This allows us to express the Apex launch cost and payload ratio as functions below:

$$\eta_{rocket2} = \frac{1}{\frac{\gamma}{1 - \eta_{rocket1}} + 1} \approx \frac{1}{1.01\gamma + 1}$$

$$C_{rocket2} \approx \gamma \cdot C_{rocket1}$$

Using Carbon Dioxide Equivalent as a unified metric and assuming Metha lox propulsion, we define a baseline constant to measure the environmental impact of a single launch event. This index consists of four dimensions:

- 1. Atmospheric Chemical Pollution:** This factor accounts for the direct emissions from methane combustion and the high-altitude Global Warming Potential (GWP) of stratospheric Black Carbon (BC), calculated as:

$$E_{atm} \approx 2,360 + (0.4 \times 4,000) \approx 4,000 \text{ tons } CO_2e.$$
- 2. Manufacturing & Resource Consumption:** This encompasses the embodied carbon from fuel production electricity (approx. 1,200 tons) and the scarcity shadow price of non-renewable Helium, resulting in a total resource cost of:

$$E_{res} \approx 1,200 + (0.1 \times 10,000) \approx 2,200 \text{ tons } CO_2e.$$
- 3. Space Debris Risk:** We estimate the economic risk of orbital debris by assuming 0.05 long-term fragments per launch with a cleanup valuation of 100 tons CO_2e per fragment.

$$E_{deb} \approx 0.05 \times 100 = 5 \text{ tons } CO_2e.$$
- 4. Local Ecological Impact:** To quantify biodiversity disruption caused by noise and chemical deposition, we introduce a sensitivity factor S_i into the model as $E_{loc,i} = S_i \times 500 \text{ tons } CO_2e$, where high-sensitivity zones like Florida wetlands are assigned $S_i = 1.5$.

5 Model Design

5.1 Phase I: Deterministic Logistics Model Design & Economic Assessment

5.1.1 Definition of Decision Variables & Construction of Multi-Objective Function

To formulate the model, we define the following decision variables:

$x_{e,t}$: Transport volume via the Space Elevator system in year t . (tons)

$y_{r,i,t}$: Transport volume via the chemical rocket system from launch site i in year t . (tons)

$u_t/w_{i,t} \in \{0, 1\}$: Binary indicator variables representing whether the transport project is active in year t .

The construction of the lunar colony involves a fundamental trade-off between the total construction duration $T_{construct}$ and the total cost C_{total} .

However, these two metrics have disparate dimensions (years vs. trillions of dollars), making a direct integrated evaluation difficult. To facilitate the solution, we employ the Weighted Sum Method to normalize these conflicting objectives into a single objective function Z . The optimization objective is formulated as:

$$\min Z = \alpha \cdot \frac{C_{total}}{C_{scale}} + (1 - \alpha) \cdot T_{construct} \quad (0.1)$$

where:

α is the weighting coefficient, representing the agency's preference between budget conservation and time-saving.

C_{scale} is a scaling factor (set to 10^{12} in this study), used to align the orders of magnitude between cost and time.

5.1.2 Scenario-based Transportation Planning and Modeling

(1) Definition of General Dynamic Factors Across Scenarios

To characterize the evolution of rocket transportation costs, we define the following parameters: let λ_0 denote the cost decay rate, and $\rho_{\min}$ represent the asymptotic lower bound factor. Let $C_{rocket1}(t)$ and $C_{rocket2}(t)$ denote the unit costs in year t for direct Earth-to-Moon delivery and delivery originating from the space elevator's anchor station, respectively. The cost dynamics are formulated as:

$$\begin{cases} C_{rocket1}(t) = \max\{\rho_{\min}, e^{-\lambda(t-1)}\} \cdot C_{rocket1} \\ C_{rocket2}(t) = \max\{\rho_{\min}, e^{-\lambda(t-1)}\} \cdot C_{rocket2} \end{cases} \quad (0.2)$$

Regarding the operational capacity of terrestrial launch sites, we introduce a growth rate λ_1 . Let $Freq_{\max,i}(t)$ be the maximum annual launch frequency for the i th launch site in year t . The calculation is given by:

$$Freq_{\max,i}(t) = (1 + \lambda_1)^{\lfloor \frac{t-1}{10} \rfloor} \cdot Freq_{\max,i} \quad (0.3)$$

(2) Scenario A: Transportation Using Only Space Elevator

The space elevator system operates as a two-stage logistics chain:

- Stage 1: Cargo is hoisted from the Earth terminal to the Apex Anchor via cable climbers.
- Stage 2: Cargo is then dispatched from the Apex Anchor to the lunar surface using transfer rockets.

Consequently, the total cost is partitioned into two components: the hoisting cost and the orbital transfer cost. Accounting for the fuel consumption overhead (represented by the coefficient $\eta_{rocket2}$), the total cost C_{total} is formulated as:

$$C_{total} = \sum_{t=1}^{T_{total}} (x_{e,t} \cdot u_t \cdot C_{elevator} + \eta_{rocket2} \cdot x_{e,t} \cdot u_t \cdot C_{rocket2}(t)) \quad (0.4)$$

The model is subject to the following constraints:

$$\begin{cases} \sum_{t=1}^{T_{max}} \eta_{rocket2} \cdot x_{e,t} \cdot u_t \geq M_{total} \\ T_{total} \leq T_{max} \\ x_{e,t} \leq 3 \cdot Cap_{elevator} \cdot u_t \\ x_{e,t} \geq 0 \end{cases}$$

(3) Scenario B: Transportation Using Only Conventional Chemical Rockets

In this scenario, cargo is deployed via chemical rockets from ten global launch sites. These sites exhibit heterogeneity in terms of both their payload capacities and their maximum sustainable annual launch frequencies (cadence). The total cost is calculated as follows:

$$C_{total} = \sum_{t=1}^{T_{total}} \sum_{i=1}^{N_{site}} y_{r,i,t} \cdot u_{i,t} \cdot C_{rocket2}(t)$$

The model is subject to the following constraints:

$$\begin{cases} \sum_{t=1}^{T_{total}} \sum_{i=1}^{N_{site}} y_{r,i,t} \cdot u_{i,t} \geq M_{total} \\ T_{total} \leq T_{max} \\ y_{r,i,t} \leq Freq_{max,i}(t) \cdot u_{i,t} \\ y_{r,i,t} \geq 0 \end{cases}$$

(4) Scenario C: Combined Use of the Two Transport Modes

When utilizing both transportation modes concurrently, the model incorporates the full set of decision variables from both systems. The objective function for the total cost is reformulated as the superposition of the costs from Scenarios A and B:

$$C_{total} = \sum_{t=1}^{T_{total}} (x_{e,t} \cdot u_t \cdot C_{elevator} + \eta_{rocket2} \cdot x_{e,t} \cdot u_t \cdot C_{rocket2}(t)) + \sum_{t=1}^{T_{total}} \sum_{i=1}^{N_{site}} y_{r,i,t} \cdot u_{i,t} \cdot C_{rocket2}(t)$$

Similarly, the mass delivery requirements are consolidated into a single global constraint:

$$\sum_{t=1}^{T_{total}} \sum_{i=1}^{N_{site}} y_{r,i,t} \cdot u_{i,t} + \sum_{t=1}^{T_{max}} \eta_{rocket2} \cdot x_{e,t} \cdot u_t \geq M_{total}$$

The remaining constraints represent the union of the constraint sets defined in Scenarios A and B.

(5) Solution Approach and Preliminary Conclusions

The formulated model is characterized as a Mixed Integer Programming (MIP) problem. By leveraging the Gurobi solver to execute a Branch-and-Bound search, we ensure the identification of global optima for timeline scheduling and cost allocation, rather than settling for local optima. This provides a rigorous mathematical foundation for our subsequent strategic recommendations.

Figure 4 illustrates the optimal objective function values across different scenarios as a function of the weighting coefficient α :

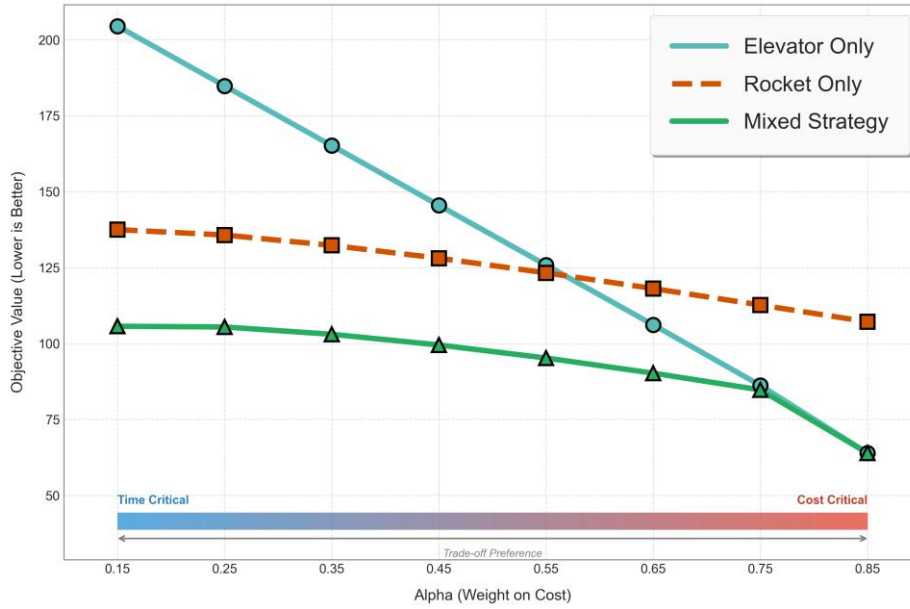


Figure 4: Strategy comparison under Alpha-weighted Cost Trade-offs

Our findings indicate that in a deterministic environment, the hybrid strategy consistently outperforms any single transportation mode. This result aligns with the Theory of Comparative Advantage in Economics: rockets possess a comparative advantage during the initial stages when the time-value of cargo is paramount; conversely, the space elevator demonstrates superior cost-efficiency for long-term, bulk-logistics missions. Consequently, the model extensions in the following stages are built upon this hybrid framework.

5.1.3 Basical Model Result

The total cost and completion time for our hybrid strategy across a range of α values are illustrated in Figure 5:

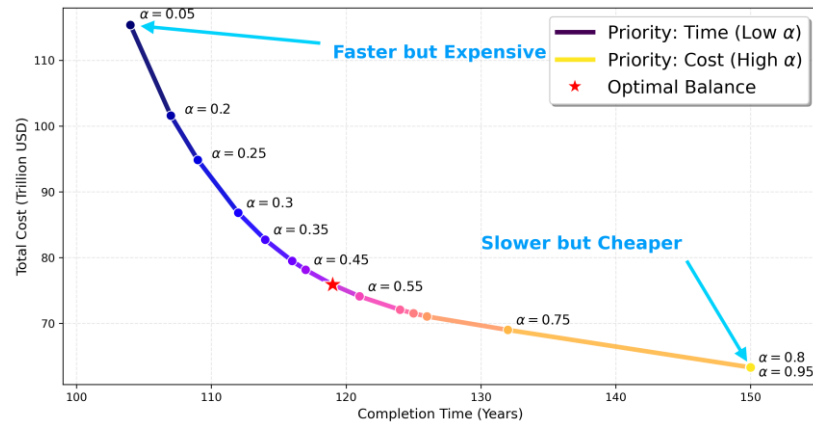


Figure 5: α -Driven Time–Cost Trade-offs

Figure 5 reveals a clear Pareto frontier, demonstrating the sensitivity of project outcomes to the weighting coefficient.

To further investigate the operational details, we examine the scenario where $\alpha = 0.5$. This value represents a balanced preference, assigning equal importance to time reduction and budget conservation. Under this specific optimization target, the temporal evolution of costs, transport volumes, and the geospatial allocation of tasks across launch sites are presented below:

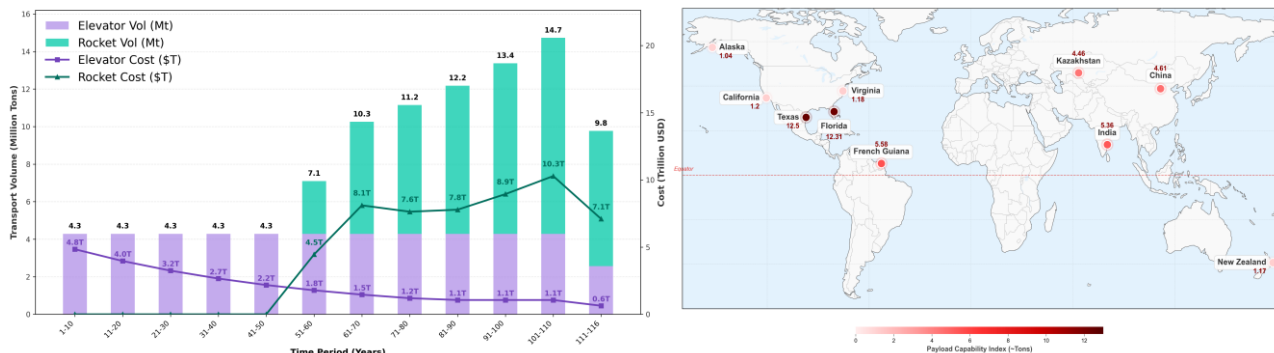


Figure 6: Temporal Logistics Distribution and Geospatial Deployment

Under this balanced configuration:

- **Temporal Evolution:** The bar chart highlights a strategic transition. In the initial phases, the space elevator handles the majority of the throughput because of the high cost of rockets. As time progresses, rocket's transport volume scales up, eventually becoming the primary mode for bulk cargo due to its superior time efficiency.
- **Geospatial Allocation:** The world map illustrates the site-specific utilization based on the heterogeneous payload capacities of the ten terrestrial launch bases. Launch sites situated at low latitudes with higher maximum launch frequencies are allocated a significantly greater aggregate throughput in the optimal solution.

5.2 Phase II: Stochastic Resilience & Adaptive Strategy

While Phase I yields a static mathematical optimum, its 'perfect world' assumption creates a **tightly coupled system lacking redundancy**.

Real-world Earth-Moon logistics face severe stochasticity, where minor faults risk cascading failures. Thus, we must shift from cost minimization to system robustness, transitioning to Phase II: Risk Modeling.

5.2.1 Mathematical Representation of Risk Features

We abstract system risks into two representative modes of supply chain disruption:

(1) Asset-Destruction Risk (Rocket Segment): A rocket launch operates as a Bernoulli Trial; failure results in simultaneous Asset Loss and Sunk Cost. This directly compromises the cumulative cargo constraint, causing the effective transport volume $S_{i,t}$ of fall short of the planned volume $N_{i,t}$. Assuming a single-launch failure probability P_{fail_rocket} , the actual number of launches follows a Binomial Distribution:

$$\left\lceil \frac{S_{i,t}}{Cap_{rocket,i}} \right\rceil \sim B\left(\left\lceil \frac{N_{i,t}}{Cap_{rocket,i}} \right\rceil, 1 - P_{fail_rocket}\right)$$

Accordingly, we define the cargo loss volume L_{rock} and sunk cost incurred by rocket C_{waste} failures in year t as:

$$L_{rock}(t) = \sum_{i=1}^{N_{site}} \left(\left\lceil \frac{N_{i,t}}{Cap_{rocket,i}} \right\rceil - \left\lceil \frac{S_{i,t}}{Cap_{rocket,i}} \right\rceil \right) \times Cap_{rocket,i}$$

$$C_{waste}(t) = \sum_{i=1}^{N_{site}} \left(\left\lceil \frac{N_{i,t}}{Cap_{rocket,i}} \right\rceil - \left\lceil \frac{S_{i,t}}{Cap_{rocket,i}} \right\rceil \right) \times C_{rocket}(t)$$

(2) Service Degradation Risk (Elevator Segment): In contrast to the binary nature of rocket launches, space elevator malfunctions—such as mechanical wear or tether oscillation—typically manifest as Capacity Derating. We introduce a time-variant Effective Capacity Factor $\eta_t \in [0, 1]$ to quantify the elevator's operational status in year t . Any reduction in this factor causes a backlog of planned cargo, resulting in subsequent schedule slippage. The Probability Mass Function of η_t is defined as:

$$\eta_t = \begin{cases} 0 & , \text{ with probability } P_{mech} \text{ (Completely Shutdown)} \\ 1 - \gamma_{sway} & , \text{ with probability } (1 - P_{mech}) \cdot P_{sway} \text{ (Capacity Reduction)} \\ 1 & , \text{ with probability } (1 - P_{mech}) \cdot (1 - P_{sway}) \text{ (Normal Operation)} \end{cases}$$

Here, P_{mech} represents the probability of Mechanical Failure, necessitating extended repairs that reduce annual capacity to zero. P_{sway} denotes the probability of tether oscillation (induced by Coriolis forces or debris), which requires operational speed reduction or pod suspension, scaling effective capacity by a factor of $1 - \gamma_{sway}$. Consequently, the actual maximum capacity ceiling for the Space Elevator in year t is revised as:

$$Cap_{ele}^{actual}(t) = Cap_{ele}^{nominal} \times \eta_t$$

These definitions translate qualitative risk descriptions into random variables operable within our mathematical framework. This formulation encompasses both **cargo loss** and **schedule delay risks**,

establishing a robust foundation for subsequent Monte Carlo stress testing.

5.2.2 Monte Carlo Stress Testing and Model Vulnerability Analysis

To quantify the operational reliability of the Phase I strategy, we developed a **Monte Carlo simulation engine**. Through $N = 1000$ independent iterations, we simulated system performance under stochastic perturbations. The results are illustrated below:

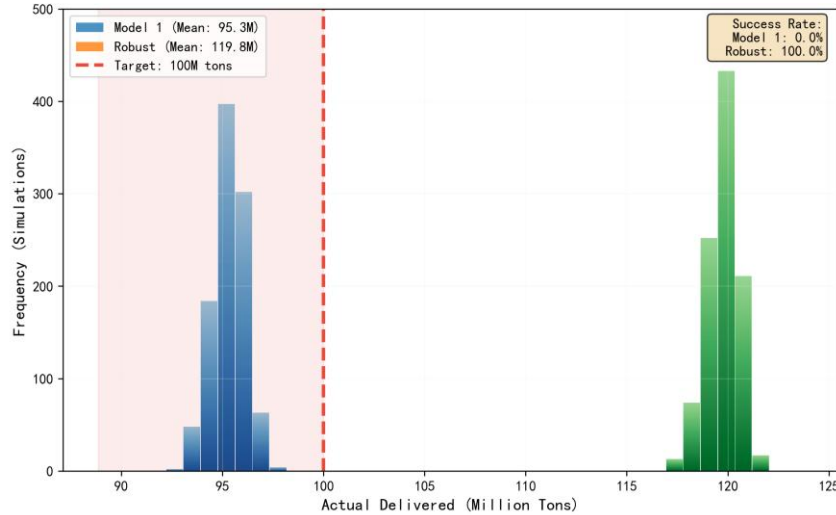


Figure 7: Distribution of Actual Deliveries

Simulation results expose the extreme fragility of the deterministic model. Upon incorporating stochastic failures, the success rate of the original optimal strategy plummets to 0%, with an average cargo shortfall of 4.68 million tons.

The root cause of this systemic failure is the Zero Redundancy design. In its drive to minimize costs, Phase I's algorithm saturated annual capacity utilization at 100%. This effectively stripped the system of any Safety Margin to buffer against stochastic perturbations.

5.2.3 Dual-Buffer Robust Optimization Strategy

Inspired by Resilient Supply Chain Management, we propose a **Dual Buffering Strategy** that proactively incurs Redundancy Costs to purchase system Certainty.

(1) Demand-Side: Safety Stock Buffer

Given the irreversibility of rocket-induced cargo loss, we introduce a safety redundancy coefficient λ_{safe} to inflate the total transport target. Analogous to Safety Stock in inventory management, this buffer offsets potential in-transit losses. Consequently, the total demand constraint is revised as:

$$\sum_{t=1}^{T_{total}} \sum_{i=1}^{N_{site}} y_{r,i,t} \cdot u_{i,t} + \sum_{t=1}^{T_{max}} \eta_{rocket2} \cdot x_{e,t} \cdot u_t \geq (1 + \lambda_{safe}) M_{total}$$

This implies that the surplus transport volume is designated to offset potential attrition caused by rocket failures.

(2) Supply-Side: Capacity Discount Factor

To account for the space elevator's operational instability, we preemptively 'discount' its nominal capacity in the planning phase. We introduce derating coefficients ξ_{ele} and ξ_{rock} , defined as:

$$\xi_{ele} = (1 - P_{mech}) \cdot (1 - P_{sway} \cdot \gamma_{sway})$$

$$\xi_{rock} = 1 - P_{fail}$$

This compels the planning algorithm to refrain from saturating nominal capacity, thereby reserving a 'Capacity Buffer' to accommodate unscheduled maintenance or reduced-speed operations. Accordingly, the revised capacity constraint is formulated as:

$$\begin{cases} x_{e,t} \leq 3 \cdot \xi_{ele} \cdot Cap_{elevator} \cdot u_t \\ y_{r,i,t} \leq \xi_{rock} \cdot Freq_{max,i}(t) \cdot u_{i,t} \end{cases}$$

5.2.4 Robustness Validation and Risk Premium

Incorporating the aforementioned probabilistic constraints, we re-solved the model and executed a subsequent Monte Carlo simulation. The results indicate:



Figure 8: Comparative Analysis: Deliveries, Cost-Time Trade-offs & Cumulative Payload Loss

- **Success Rate Resurgence:** The system's mission success rate across all simulated failure permutations rebounded to 100%.
- **Quantification of Risk Premium:** This reliability entails a cost. Total expenditure escalated from \$68.3 Trillion to \$86.3 Trillion (+26.4%). This incremental \$18 trillion should not be

viewed as waste, but rather as a necessary Insurance Premium to safeguard this megaproject against catastrophic collapse.

5.2.5 Adaptive Substitution Mechanism under Extreme Scenarios

To examine the dynamic adaptation logic of the Earth-Moon logistics system under extreme conditions, we formulated four representative stress scenarios. Holding the subjective decision preference constant at $\alpha = 0.5$, we contrasted the Transport Composition across distinct risk profiles. This analysis elucidates the **adaptive substitution mechanisms** intrinsic to the logistics network:

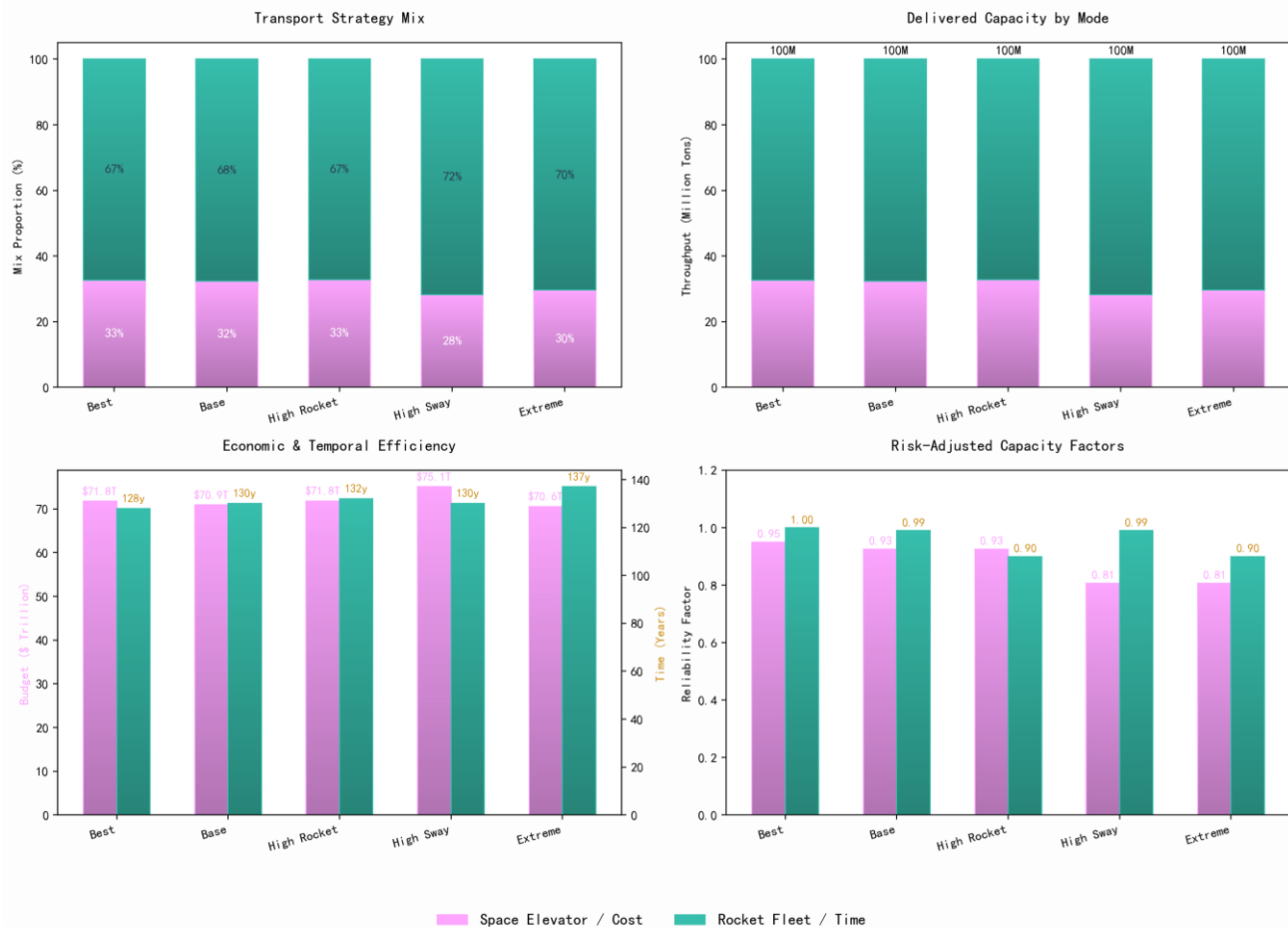


Figure 9: Multi-Scenario Evaluation of Space Elevator and Rocket Fleet Strategies

Figure 9 reveals three critical shifts in system behavior under varying risk profiles:

1. Risk Aversion under High Rocket Failure

When rocket failure rates escalate to 10%, the scheduling logic prioritizes **high-certainty delivery** over time sensitivity. The system fully exploits the space elevator's capacity to circumvent payload loss, deliberately extending the total project make span to hedge against the prohibitive risks of rocketry.

2. Forced Aggression under High Sway Risk

Severe tether sway represents the most challenging scenario, dropping elevator utilization to 28%. To bridge the resulting "temporal void," the model pivots from cost-efficiency to **schedule-preservation**, drastically increasing rocket frequency. This "forced structural drift" ensures mission

completion but inevitably precipitates a significant surge in expenditure.

3. Asymmetry of Substitution Effect

A critical finding is the **asymmetry** between modes. Rocket substitution is "unbounded" (financially scalable), whereas elevator substitution is "bounded" (physically capped at 537,000 tons/year). Consequently, enhancing elevator stability takes strategic precedence over rocket reserves. The "Extreme Risk" scenario incurs a **7.6% cost premium** (\$70.9T to \$76.3T), quantifying the price of logistical resilience.

This scenario-based analysis demonstrates the model's sensitivity to minute environmental shifts and its ability to generate logically self-consistent resource reorganization schemes, providing a scientific basis endowed with Strategic Flexibility for lunar base construction.

5.3 Phase III: Long-term Metabolic Supply Chain & Inventory Management

While Phases I and II concentrated on the capital-intensive demands of **Construction Logistics**, Phase III transitions to **Operational Logistics**, supporting a mature settlement of 100,000 inhabitants. This shift marks a fundamental evolution from a **Project Management** paradigm—prioritizing makespan minimization—to an **operations management** framework, designed to minimize OPEX while maximizing service levels.

5.3.1 Analysis of Annual Total Water Demand for a Lunar Colony

To achieve '**Systemic Equilibrium**' in a mature lunar metropolis, annual replenishment M_{water} must offset irreducible losses across three sectors: Domestic D_{dom} , Agricultural D_{agr} , and Industrial D_{ind} .

Based on FAO/World Bank data and NASA's ECLSS 99% recycling efficiency, domestic demand for a population $P = 10^5$ is: $D_{dom} = P \times 66.49 \times (1 - 0.99) = 6.649 \times 10^4$ t/a.

Agricultural demand in the Controlled Ecological Life Support System (CELSS) stems from condensation recovery inefficiencies and 'biological lock-in' (water assimilated into biomass). The model is defined as:

$$W_{Makeup_Agr} = (A \cdot E_{tr} \cdot 365 \cdot (1 - \eta_{agr})) + (P \cdot m_{food_water} \cdot 365).$$

The total annual agricultural water replenishment is calculated based on the total planting area A , (based on standards from NASA) the daily plant transpiration rate E_{tr} , the condensation recovery efficiency η_{agr} , and the biological lock-in volume m_{food_water} derived from dietary structures.

Industrial makeup water offsets electrolysis-based oxygen generation, thermal control leakage, and maintenance losses: $W_{Makeup_Ind} = (L_{O2} \cdot 1.125) + (Q_{thermal} \cdot r_{leak}) + W_{maint}$.

Aggregating these sectors, the total annual water replenishment requirement for a lunar colony of 100,000 inhabitants M_{water} is determined to be 116,460 tons/year.

5.3.2 Supply Channel Selection and Economic Evaluation

During the operational phase, cargo—predominantly water—is characterized by its bulk volume, high predictability, and low time-sensitivity. We conducted a Comparative Cost Analysis to identify the most economically viable transport mode.

We contrasted the Annual Operating Expenses (OPEX) of a rocket-only resupply strategy against a space elevator-based approach:

$$WC_{rocket} = M_{water} \times C_{rocket1} \approx 573.8 \text{ Billion/year}$$

$$WC_{ele} = M_{water} \times C_{elevator} \approx 0.155 \text{ Billion/year}$$

Our analysis reveals that the **marginal transport cost** of the space elevator is merely **0.027%** of that associated with traditional rocketry. Furthermore, from the perspective of **Sustainability Economics**, the space elevator is not merely an option, but the **sole financially viable pathway** for sustaining life-support logistics. Consequently, our model designates the space elevator as the primary 'Pipeline' for routine replenishment, while relegating rockets to the role of a '**Strategic Reserve**' for emergency contingencies.

5.3.3 Inventory Dynamics & Schedule Optimization

To operationalize the supply chain, we applied **Inventory Control Theory**. We constructed a discrete-time inventory model to determine the **optimal Delivery Schedule** and safety stock levels. The framework is established as follows:

(1) Inventory Balance Equation

Let I_t denote the lunar water inventory level at the end of month t . The replenishment dynamics are governed by:

$$I_t = I_{t-1} + A_t - C_t$$

Where A_t denotes the total volume of water arriving on the Moon in month t , and C_t represents the constant monthly water consumption.

(2) Safety Buffer Constraints

Integrating the risk assessment from Phase II, we introduce Service Level Constraints. Given that Tether Sway could precipitate a service interruption of approximately one month, we established the following boundary conditions S_{safe} :

- **Safety Stock Threshold:** $I_t \geq C_t \approx 10000$ tons. This redundancy guarantees that even in the event of a 30-day elevator outage, the colony can sustain its fundamental metabolic requirements.
- **Storage Capacity Upper Limit:** The maximum physical capacity of the lunar water storage facility is constrained to $S_{max} = 20,000$ tons.

5.3.4 Optimized Resupply Schedule and Strategic Analysis

Leveraging **Gurobi optimization**, we formulated a comprehensive annual 'Logistics Schedule' for the lunar colony. To achieve load smoothing across the Earth-Moon logistics network, we propose the '**Weekly Pulse Strategy**'.

1. Optimal Schedule

- **Replenishment Frequency:** Maintained as a routine weekly docking event.
- **Single Mission Payload:** Approximately 2,240 tons per shipment (equating to a monthly delivery volume of 8,960 - 11,200 tons).

This replenishment intensity utilizes merely 21.6% of the space elevator's total monthly capacity (based on an annual throughput of 537,000 tons). The remaining capacity can robustly support the daily passenger transit and industrial expansion required for a population of 100,000.

2. Visualization Analysis: Saw-tooth Dynamics

As illustrated in Figure 10, the lunar base's inventory level exhibits a classic **Saw-tooth Pattern**. At each replenishment node, the inventory level surges instantaneously upon cargo arrival, followed by a gradual linear decline due to the metabolic consumption of the inhabitants. Because the SSRIM model incorporates the risk buffers defined in Phase II, the inventory trajectory consistently operates **above the safety red line** of 1.46×10^4 tons.

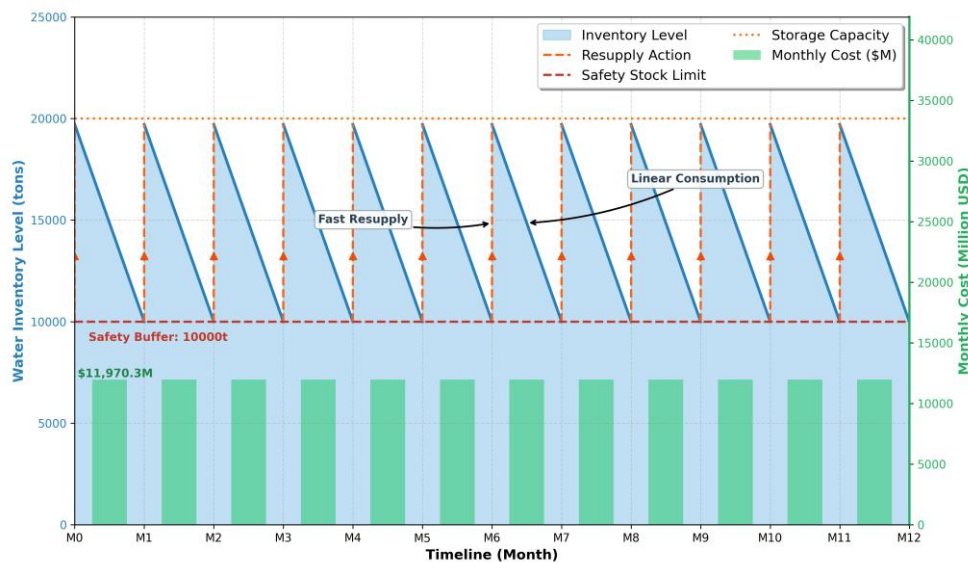


Figure 10: Dynamic Water Inventory Management with Resupply and Cost Over Time

3. Strategic Insights: Long-term Operations Assessment

The analysis demonstrates that leveraging the **LUNALINK** system keeps the cost of maintaining the 'respiration and metabolism' of a 100,000-person city (at \$155 million annually) negligible and controllable within the lunar colony's economic structure. This characteristic of '**High**

CAPEX, Low OPEX' further validates the strategic value of the space elevator as the fundamental artery between Earth and the Moon.

5.4 Phase IV: Environmental Exteriority & Green Evaluation

With the logistics network established, we proceed to address the 'hidden costs' of space exploration. Our objective is to **internalize negative externalities**—specifically pollution and debris—into the decision-making framework, thereby catalyzing a paradigm shift from 'Cost Minimization' to '**ESG (Environmental, Social, and Governance) Optimization**'.

5.4.1 3D Multi-objective Optimization Strategy

To quantify the environmental impact, we defined four baseline indices per launch: **Atmospheric Damage** E_{atm} , **Resource Depletion** E_{res} , **Risk Equivalent** E_{deb} and **Local Ecological Impact** $E_{loc,i}$. The numerical distribution of these indices is presented below:

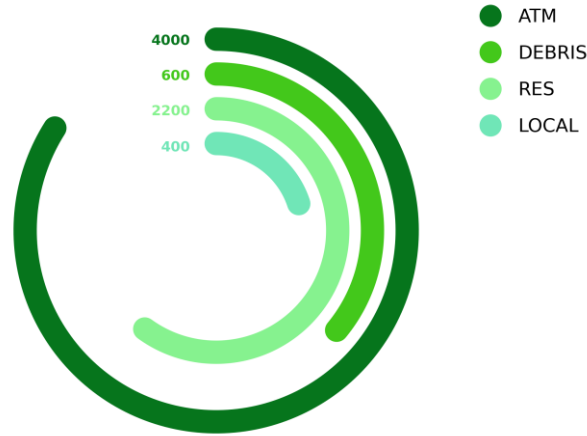


Figure 11: Environmental Impact: Four Baseline Indices with Distribution

We integrated the Environmental Impact Index (EII) into the objective function. To maintain decision neutrality and ensure consistency and comparability with previous phases, we assumed equal importance for **Time efficiency** and **Economic Cost** ($\alpha = \beta$), thereby concentrating variability solely on the environmental weight, θ . Consequently, the new objective function is redefined as:

$$\min Z = \alpha \cdot \frac{Cost}{Ref_C} + \beta \cdot \frac{Time}{Ref_T} + \theta \cdot \frac{EII}{Ref_E}$$

The constraint $\alpha = \beta = (1 - \theta)/2$ isolates the environmental variable, enabling us to clearly observe the system's structural realignment of transport resources as θ increases.

5.4.2 Analysis of Environmental Trade-offs

We conducted a sensitivity analysis on θ to delineate the **Pareto Frontier** of the Earth-Moon logistics system. The results are illustrated in the figure below:

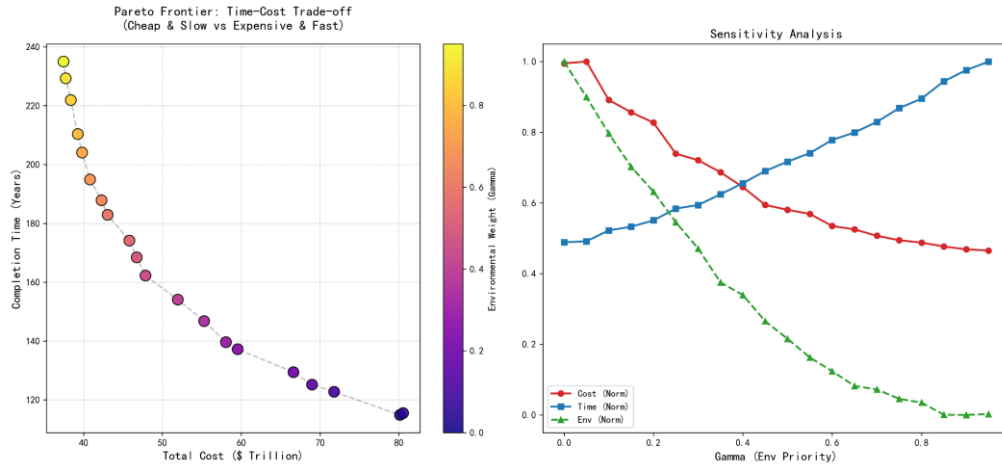


Figure 12: Time-Cost Pareto Frontier with Environmental Priority Sensitivity

Sensitivity analysis of the environmental weight θ reveals a Pareto Frontier defining the trade-off between high-cost, rapid rocket-based deployment $\theta \rightarrow 0$ and low-cost, eco-friendly space elevator logistics $\theta \rightarrow 1$.

The system exhibits a critical strategic fulcrum at $\theta \approx 0.4$. Below this threshold, heavy-lift rockets maintain construction intensity; above it, the space elevator achieves dominance. Paradoxically, the "Eco-friendly Scenario" proves more financially economical in the long run due to the elevator's superior unit-cost efficiency. While the Environmental Impact Index (EII) drops precipitously for $\theta \in [0, 0.4]$, further increases lead to diminishing returns and prohibitive temporal costs—exceeding 220 years for $\theta > 0.8$.

Strategic Recommendation: We propose an optimal decision interval of $\theta \in [0.35, 0.45]$. This range suppresses environmental impact to below 40% of the baseline while maintaining a balanced construction cycle (~140-150 years).

Ultimately, the LUNALINK system functions as a self-regulating framework, enabling precise calibration of "greenness" to realize Sustainable Interstellar Civilization Governance.

6 Sensitivity Analysis

6.1 Global Parameter Screening: The Morris Method Approach

To navigate the high-dimensional uncertainties of lunar colonization, we employ the **Morris Elementary Effects Method** to screen parameters based on their influence magnitude μ^* and degree of non-linear interaction σ .

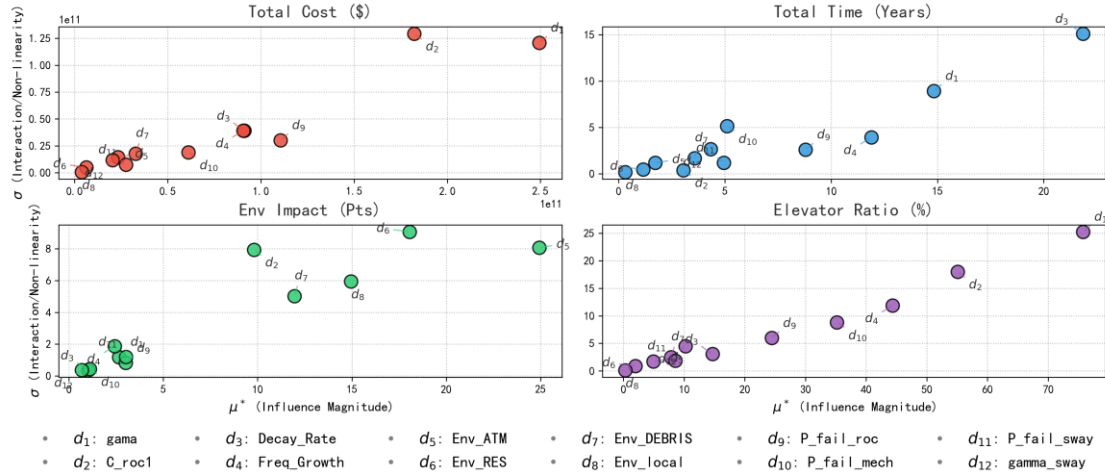


Figure 13: Multi-Dimensional Pareto Analysis of Time, Cost, and Environmental Impact

Key findings across the objective dimensions include:

- **Cost & Decision-Making:** Rocket pricing $C_{rocket1}$ and the policy-weighting factor γ emerge as the most profound drivers. Their location in the high- μ^*/σ quadrant indicates that project costs and payload allocation strategies are governed by complex, non-linear interactions between these factors.
- **Time Sensitivity:** The cost decay rate is the dominant determinant for project duration. This highlights technological maturation—manifested as accelerated cost reduction—as the primary catalyst for schedule compression.
- **Environmental Impact:** A market-driven feedback loop is observed; lower rocket costs incentivize intensive deployment, thereby exacerbating atmospheric degradation and energy consumption. Conversely, higher costs act as an implicit constraint that promotes environmental sustainability.

In summary, $C_{rocket1}$ and γ serve as the fundamental anchors for system optimization and strategic planning.

6.2 Quantitative Local Sensitivity Analysis

Building on the Morris method, a One-at-a-Time perturbation ($\pm 20\%$) was employed to quantify the local impacts of key parameters. The resulting Tornado Diagrams (Figure 14) rank the influence of these variables on timeline, cost, environmental footprint, and transportation structure.

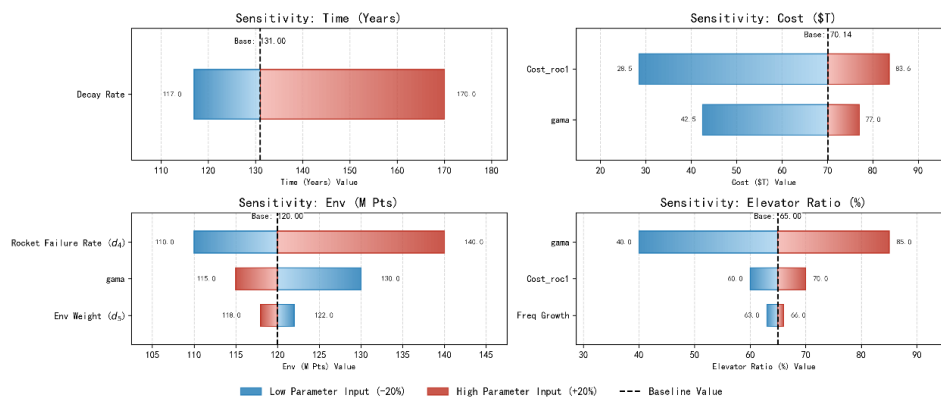


Figure 14: Sensitivity and Trade-off Analysis of Time, Cost, and Env Impact

(1) Impact of Technological Iteration on Construction Timeline

Project duration is critically sensitive to the **Tech Decay Rate** d_3 . A $\pm 20\%$ variance in d_3 yields a 53-year swing (117–170 years) relative to the 131-year baseline. This "Technology Dependency Trap" underscores that, under fixed capital investment, the iteration speed of rocket cost-efficiency constitutes the primary scheduling bottleneck.

(2) Rocket-Cost-Driven Budgetary Framework

Rocket Unit Cost d_2 emerges as the dominant budgetary driver. A 20% cost reduction precipitates a 60% budget decrease (from \$70.14T to \$28.5T), while a 20% overrun elevates it to \$83.6T. This high leverage stems from the heavy front-loaded reliance on chemical rockets, making launch cost mitigation (e.g., via reusability) a financial imperative.

(3) Reliability-Driven Environmental Footprint

Counter-intuitively, the **Rocket Failure Rate** d_4 is a more significant environmental determinant than the weighting coefficient d_5 . A $\pm 20\%$ fluctuation in d_4 shifts the Environmental Impact Index between 110 and 140 M Pts. This validates a "Safety is Green" paradigm, where enhancing system reliability is more effective for ecological protection than policy-based penalties.

(4) Highly Sensitive Transport Structure

The parameter γ (d_1) governs the system's "morphology." A 20% improvement in γ increases the space elevator's cargo share from 65% to 85%; conversely, underperformance drops its share to 40%. This high "Structural Elasticity" allows the LUNALINK project to pivot between rocket-centric and elevator-centric pathways based on technological maturity.

7 Strengths and Weaknesses

7.1 Strengths

- Our LUNALINK strategy provides a holistic logistics roadmap, spanning from initial infrastructure construction to long-term metabolic maintenance (water resupply). It bridges the gap between macro-level transportation planning and micro-level inventory management.

- Our strategy is risk-awareness. By simulating stochastic failures—such as rocket launch anomalies and space elevator oscillations—through Monte Carlo stress testing, it's proved that our model can still provide valuable strategies under the impact of certain external risks..
- Our model internalizes environmental externalities (e.g., atmospheric damage, space debris) into the objective function. This allows for a 3D multi-objective optimization that balances economic cost, time efficiency, and "Green" interstellar governance.
- The "Weekly Pulse Strategy" offers a concrete, discretized timetable for water delivery. By utilizing inventory control theory, the model guarantees the sufficient water of 100,000 inhabitants even during extended elevator outages, making the plan practically viable.

7.2 Weaknesses

- Our model relies on the extrapolation of current technological trends to the year 2050 and beyond. Factors such as the cost decay rate and $C_{rocket2}$ are based on theoretical assumptions that may be subject to unpredictable paradigm shifts in aerospace engineering. But this is unavoidable when predicting future scenarios.
- We simplified the orbital dynamics. To maintain computational feasibility over a century-long horizon, the model utilizes linearized approximations for complex orbital transfer scenarios and fuel consumption ratios. This may omit higher-order perturbations in Earth-Moon logistics.

8 Conclusion

In this study, we developed the LUNALINK framework, a systemic solution for establishing and sustaining a 100,000-person lunar colony. Our analysis demonstrates that a single-mode transportation system is insufficient for the demands of lunar colonization. Instead, a Rocket-Space Elevator Hybrid Strategy provides the optimal balance, leveraging the immediacy of rockets for early-stage construction and the cost-efficiency of elevators for bulk operational logistics.

Key findings indicate that while risk-blind deterministic plans suffer from a 0% success rate under stochastic failures, our Dual-Buffer approach restores reliability to 100% by introducing strategic safety stocks and capacity derating. Furthermore, for long-term operations, the space elevator proves to be the only financially viable "artery," reducing water resupply costs by over 99.9% compared to traditional rocketry.

Finally, by applying the Economic-Efficiency-Green-Evaluation model, we identified a Pareto optimal decision interval for environmental weighting. This suggests that the transition to a sustainable interstellar civilization requires a deliberate shift from pure cost minimization to a balanced governance model that respects the terrestrial and lunar ecological footprints. The LUNALINK

framework provides a robust, quantifiable, and environmentally responsible blueprint for humanity's expansion into the cosmos.

MEMORANDUM

To: The Director of the Moon Colony Management (MCM) Organization

From: Team 2601727

Date: January 28, 2026

Subject: Strategic Roadmap for Establishing and Sustaining a 100,000-Person Lunar Colony

Distinguished Members,

Based on our simulation of the Earth-Moon Logistics Network, we propose the "LUNALINK" strategy. Our research indicates that only a systemic approach integrating **"Hybrid Transportation, Resilient Redundancy, and Green Sustainability"** can ensure absolute mission success and economic feasibility over a construction horizon spanning more than a century.

I. Construction: Hybrid Drive & "Dual-Buffer" Resilience

During the infrastructure construction phase, we strongly recommend abandoning a single-mode transportation model in favor of a **Rocket-Space Elevator Hybrid Strategy**, leveraging rocket immediacy for early phases and elevator cost-efficiency for bulk transport. Static plans yield a **0% success rate** under risk; thus, we propose a **"Dual-Buffer Robust Strategy"** featuring demand-side safety stocks and supply-side capacity discounts.

1. **Demand-Side:** Introduction of a safety stock buffer to offset potential asset losses resulting from rocket launch failures.
2. **Supply-Side:** Application of a capacity discount factor during planning to accommodate service degradation caused by tether sway in the space elevator system.

While this raises the budget to **\$86.3 trillion** (a 26.4% premium) and extends the duration to **147 years**, it restores success rates to **100%**. An adaptive mechanism is also required to dynamically switch modes during failure events.

II. Operations: Metabolic Balance & Pulse Resupply

Sustaining the colony requires **116,460 tons** of water annually. Rocket resupply would cost **\$573.8 billion/year**, whereas the space elevator costs only **\$0.155 billion**—making it the only viable pillar. We recommend a **"Weekly Pulse Strategy"** (~2,240 tons/week) to maintain inventory above the **14,558-ton safety line**, guaranteeing survival during 45-day elevator outages.

III. Environmental Responsibility

In the pursuit of interstellar expansion, we must not ignore the negative externalities imposed on the terrestrial environment. Using our EEGE model, we advise setting the environmental weight coefficient γ to **[0.35, 0.45]**. This interval balances a feasible **140–150-year** cycle with a **>60% reduction** in pollution and space debris risks. This represents the "Pareto Optimal" path between engineering efficiency and ecological ethics.

In conclusion, the construction of a lunar colony should not merely be a race for speed, but a long-term investment in system robustness and sustainability. The **LUNALINK** framework provides a quantifiable, risk-resistant, and environmentally responsible blueprint for implementation.

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Report on Use of AI Tools

1. Gemini 3 Pro (Jan 31, 2026 and Feb 1, 2026 version)

Purpose: Data Collection

Query: "Please collect relevant data on Falcon rockets and provide URLs; collect global water usage statistics and provide relevant data sources."

Output: The model provided a list of web URLs and descriptions of potential data sources regarding SpaceX Falcon rocket specifications and global water consumption trends. We verified the accuracy of these links manually and downloaded the verified data for further analysis.

2. Gemini 3 Pro (Feb 2, 2026 version)

Purpose: Translation and Language Polishing

Usage: Utilized these tools to translate our original draft written in Chinese into English and to refine the grammatical structure of the final report.